

Lucas LALANDE, Ph. D.

Teachings ~ 600h

Qualification sections 66, 67, 68

I am an evolutionary ecology and physiology Ph.D. willing to investigate how physiological mechanisms and evolutionary processes interact and shape each other by combining population ecology, physiological, statistical and longitudinal approaches.



Contacts

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Grenoble, France

32 years old



Languages & certifications

- **French:** native / **English:** fluent
- **Animal experimentation** (Designer level)
- **Formation in bioinformatics and genomics**
- **1 Star Diver** (Supervised diver to 20m)
- **Driving licence**

Education

Oct. 2020-Dec. 2023

Ph.D. - Ecophysiology

Stress, health and senescence in a long-lived mammal, the roe deer (*Capreolus capreolus*)
LBBE UMR 5558, Lyon, France

Sept. 2018-Sept. 2020

MSc - Behavioural Ecology and Wildlife Management

UBFC, Dijon, France

Sept. 2017-Sept. 2018

BSc - Biology - Ecology

USMB, Chambéry, France

Sept. 2012-Sept. 2015

BSc - Biology

UGA, Grenoble, France

- Sept. 2014-Aug. 2015

University of Montreal, Canada



Research & experience

Publications (4)

- Perez, G.*, Lalande L. D.* et al. **2026**. Possible exposure to unidentified coronaviruses in roe deer (*Capreolus capreolus*) populations suggested by SARS-CoV-2 serological investigation in France. Royal Society Open Science. *shared first authorship
- Lalande, L. D. et al. **2024**. Early-life glucocorticoids accelerate lymphocyte count senescence in roe deer. General and Comparative Endocrinology.
- Lalande, L. D. et al. **2023**. Glucocorticoids negatively relate to body mass on the short-term in a free-ranging ungulate. Oikos.
- Lalande, L. D. et al. **2022**. Sex-specific body mass ageing trajectories in adult Asian elephants. Journal of Evolutionary Biology.

Manuscripts in preparation (1)

- Lalande, L. D. et al. Environment-dependent haematological senescence and effects of glucocorticoids in two wild roe deer populations. **In prep.**

Research experiences

- **Postdoc Oct. 2023 – Dec. 2023:** SARS-CoV-2 in French deer species – LabEx ECOFECT, Comue Université de Lyon
- **Ph. D. Oct. 2020 – Sep. 2023:** Stress, Health and Senescence in a long-lived mammal, the Roe deer (*Capreolus capreolus*) – LBBE UMR5558, Lyon
- **M2 Internship Feb. 2020 – Aug. 2020:** Body mass senescence in Asian elephants (*Elephas maximus*) – Turku University, Finland
- **M1 Internship Mar. 2019 – May 2019:** Population variability in antipredator behaviours in *Gammarus fossarum* – Biogéosciences UMR 6282, Dijon



Teachings (~600h)

University teaching position, Sept. 2024–Sept. 2025: Biological sciences, bachelor degree – INU Champollion, Albi (**400.5 h**)

• Licence 1

Animal biology (6h CM)

Organisms and environment relationships (12h TD)

Mathematics & Introduction to R (48h TD)

Plant physiology (14h CM, 36h TD)

• Licence 2

Plant physiology & biology (16h TD, 20h TP)

Ecology (18h CM, 10h TD) – Responsible

Ethology (12h CM)

Biostatistics & R (10h CM, 28h TD, 20h TP) – Responsible

• Licence 3

Phylogeny (30h TP)

Field studies (11h CM, 6h TD, 48h TP)

R applied to ecology (8h TP)

Ph. D., Sept. 2020–Dec. 2023, Université Claude Bernard Lyon 1 (192 h)

• BUT 1 Génie Biologique

Biostatistics & R (53h TD, 86h TP)

Animal biology (24h TP)

Communication, transversal teachings (31h TD)

References

Emmanuelle Gilot-Fromont, Professor
emmanuelle.gilotfromont@vetagro-sup.fr
Ph.D. supervisor

Pauline Vuarin, Assistant Professor
pauline.vuarin@univ-lyon1.fr
Ph.D. supervisor

Jean-François Lemaître, Professor
jean-francois.Lemaitre@univ-lyon1.fr
LBBE Biodemography group leader

Virpi Lummaa, Professor
virpi.lummaa@utu.fi
Master internship supervisor

Pierre Marty
pierre.marty@univ-jfc.fr
Colleague and INU Champollion Bachelor degree responsible

Arnaud Billet
arnaud.billet@univ-jfc.fr
Colleague and INU Champollion Bachelor 2 responsible

Hobbies

- Music, Guitar
Check out our band: **Bed Market**



- Climbing, bouldering
- Chess



Resources & Packages (2)

- Lalande, L. D. **2026**. selectools : A facilitating model selection procedure (Version 0.3.0) [R]. [10.5281/zenodo.20378446](https://doi.org/10.5281/zenodo.20378446) ([Source code](#))
- Lalande, L. D. **2026**. conwaysGoL: An easy-to-handle Conway's Game of Life simulator for R (Version 0.1.1) [R]. [10.5281/zenodo.18290213](https://doi.org/10.5281/zenodo.18290213) ([Source code](#))



Skills

- **Data analyses (R):** Manipulation of large datasets, (Generalized) Linear (Mixed) Models, threshold models, Information criteria for model selection, Path modelling, Leave-One-Out Cross-Validation
- **Wildlife handling:** Roe deer capture and restraint, morphological and biological sampling (blood, faeces) as part of a CMR program in two wild populations
- **Field work:** CMR, observation, telemetry, *Gammarus fossarum* collection in natural habitat
- **Laboratory work:** breeding and behavioural assay in laboratory-controlled environment, faecal metabolite extraction, haemagglutination-haemolysis assay, ELISA assay, mouse digestive and respiratory system dissection
- **Programming:** R Package development, Nextflow, Python, Bash
- **Bioinformatics:** genomics and transcriptomics, NGS data analyses, QC, alignment, variant calling, visualization, splicing quantification, differential analyses, genome browser
- **Transparent & reproducible research:** version-control, literature management - Git/GitHub, Zotero, Obsidian
- **SQL (MySQL):** Extract and manipulate data from a large database ([Myanmar Timber Elephant Project](#))



Conferences

- **Ph.D. defense**, Lyon – Dec. 2023
Stress, health and senescence in a long-lived mammal, the roe deer
Jury: Michaela Hau (Prof., Max Planck Institute – Reviewer), Jean-François Le Galliard (Prof., Université de la Sorbonne – Reviewer), François-Xavier Dechaume-Moncharmont (Prof., Université de Lyon – Examiner), Karine Chalvet-Monfray (Prof., VetAgro Sup – Examiner), Emmanuelle Gilot-Fromont (Prof., VetAgro Sup – Ph.D. director), Pauline Vuarin (Lecturer, Université de Lyon – Ph.D. co-director)
- **EMPSEB 28**, Scotland – Jun. 2023
Oral presentation: *Is stress impacting the way you get old?*
- **Ecology & Behaviour**, Strasbourg – Mar. 2022
Poster: *The grass is always greener on the other side my deer*
- **Decryp'thèse**, Lyon – May 2022
Oral presentation: *Sex-specific body mass ageing in Asian elephants*
- **BES**, Liverpool – Dec. 2021
Poster: *The grass is always greener on the other side my deer*